Lab Manual 2: Verification of Thévenin/Norton Theorem

2.1 Introduction and learning objectives

In electrical engineering, simplifying complex circuits is essential for efficient analysis and design. Thévenin's Theorem and Norton's Theorem are powerful tools that allow engineers to reduce a network of resistors and sources to a simple equivalent circuit. Thévenin's Theorem replaces a network with a single voltage source in series with a resistance, while Norton's Theorem uses a current source in parallel with a resistance.

This lab aims to experimentally verify both theorems by constructing a resistive DC circuit, determining its Thévenin and Norton equivalents, and comparing the behaviour of the original and simplified circuits when connected to a load. Through this process, students will gain practical insight into circuit simplification techniques and validate theoretical predictions with real-world measurements.

By the end of this lab, students will be able to:

1. **Construct and analyze** a resistive DC circuit with multiple voltage sources.
2. **Troubleshoot** basic resistive circuits.
3. **Determine** Thévenin and Norton equivalents circuits.
4. **Compare** original and equivalent circuits using circuit analysis theorems and techniques.
5. **Accurately measure** voltage, current, and resistance.

2.2 Tools and Components Needed

Item	*Specification*
Resistors	R1 = 470 Ω, R2 = 820 Ω, R3 = 270 Ω, R4 = 3.9 kΩ, R5 = 560 Ω, RL = 680 Ω and 1.5 kΩ
Voltage source	12 V and 5 V DC Regulated Power Supply
Breadboard	Standard
Connecting Wires	Assorted
Measurement	Digital multimeter

2.3 Experimental Work

1. Always ensure the following safety checks are adhered:
   a. Power Off the supply before assembling the circuit or make changes to the circuit.
   b. Double-check connections to avoid short circuits.
   c. Use correct resistor values to prevent overheating.
   d. Avoid touching live wires while the circuit is powered.
   e. Handle multimeter probes carefully to avoid damage or incorrect readings.

2. Construct the circuit diagram as shown in Figure 2.1. **(DO NOT TURN ON THE DC REGULATED POWER SUPPLIES FIRST)**.

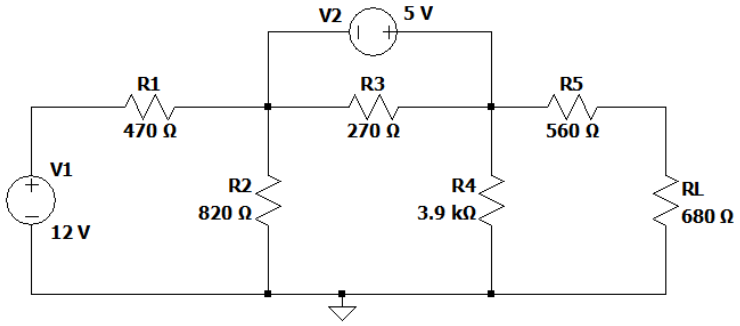


Figure 2.1

3. Verify that all connections match the circuit diagram, ask the lab executive or lecturer to check your circuit before turning on the DC Regulated Power Supply. **Take a photo of the circuit and include it in the report.**

4. Turn on the DC Regulated Power Supply, then use the digital multimeter to measure the voltage across RL and the current through RL.

5. Turn off the DC Regulated Power Supply, then change the RL to 1.5 kΩ, turn on the supply again and repeat step 4.

6. Next, remove RL and measure the terminal's open circuit voltage (V_{oc}) and the short circuit current (I_{sc}), as shown if Figure 2.2.

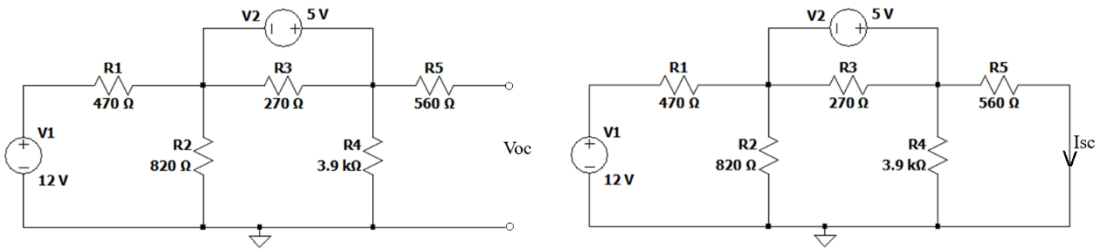


Figure 2.2

7. Lastly, disconnect all voltage supplies and replace them with wires, use the digital multimeter to measure the resistance across the open terminals (where RL was removed).
8. Record the values in Table 2.1 in the Results and Discussion section and perform calculations to verify the circuit using Thévenin/Norton Theorem.

2.4 Results and Discussion

2.4.1 Measurement data

Table 2.1

Component	Measured Voltage (V)	Measured Current (A)	Measured resistance (Ω)
RL = 680 Ω			
RL = 1.5 kΩ			
RL removed			

2.4.2 Verification of circuit using Thévenin/Norton Theorem

1. With RL removed, use any circuit analysis theorems and techniques to construct the Thévenin equivalent circuit **(show all steps and diagrams of simplifying the original circuit diagram down to the Thévenin equivalent circuit).**
2. Calculate the Thévenin voltage (V_t), the Norton current (I_n), and Thévenin resistance (R_t), **show all working steps.** Record the values in Table 2.2.

   Table 2.2

Calculated V_t (V)	Calculated I_n (A)	Calculated R_t (Ω)

3. Then, insert the RL (680 Ω or 1.5 kΩ) back to the Thévenin equivalent circuit, and calculate the voltage across and current through the respective RLs. Record the values in Table 2.3.

Table 2.3

Component	Calculated Voltage (V)	Calculated Current (A)
RL = 680 Ω		
RL = 1.5 kΩ		

4. Compare the calculated values with the measured ones. Is there discrepancy between the results? Discuss.
5. How is the Thévenin/Norton equivalents useful in circuit analysis?

Report Marking Rubric

Criteria	Poor (1 mark)	Fair (2 marks)	Good (3 marks)	Very Good (4 marks)	Excellent (5 marks)
1. Introduction & Objectives (10%)	Vague or missing; no clear objectives	Basic overview; objectives unclear or incomplete	Adequate summary; some objectives stated	Clear introduction; most objectives covered	Comprehensive intro; all objectives clearly stated and contextualized
2. Methodology: Circuit Assembly, Safety & Measurement Technique (20%)	Circuit not assembled correctly; safety rules ignored; incorrect use of multimeter	Major errors in assembly; minimal safety adherence; some correct measurements	Circuit mostly correct; some safety steps followed; mostly correct technique	Circuit correctly assembled; most safety rules followed; reliable data with few errors	Circuit perfectly assembled; all safety protocols strictly followed; flawless technique and consistent measurements
3. Results: Data Recording & Tables (20%)	Tables missing or incomplete; data unclear	Tables present but poorly formatted or inaccurate	Tables mostly complete; some formatting or data issues	Well-organized tables; mostly accurate data	Clear, complete, and accurate tables with proper formatting
4. Results: Theoretical Calculations, Rule Application, Equivalent Circuit Construction (20%)	No calculations or major errors; no working shown; rules not applied; incorrect or missing equivalents	Attempted calculations; many errors; incomplete working; unclear rule use; unclear circuit construction	Correct approach; some errors in calculation or logic; basic rule application or equivalent circuits	Mostly accurate calculations; clear working steps; rules applied with minor errors; good circuit construction	Fully accurate calculations; all steps clearly shown and justified; rules applied accurately with clear circuit diagrams; equivalents constructed correctly
5. Discussions: Comparison, Discussion & Interpretation of Results (20%)	No comparison or discussion; lacks understanding	Minimal comparison; limited insight	Some comparison; basic discussion of discrepancies	Good comparison; thoughtful discussion of results	Thorough comparison; deep, critical analysis and excellent interpretation of discrepancies and implications
6. Conclusion, Report Structure & Presentation (10%)	No sensible conclusion. Disorganized; poor formatting; hard to follow	A conclusion is not supported by any experimental evidence. Some structure; inconsistent formatting	A conclusion is only supported by some experimental evidence. Organized; readable with minor formatting issues	A conclusion is mostly supported by experimental evidence. Well-structured; clear formatting and flow	The aim of the experiment has been achieved with evidence supported conclusion. Professionally presented; logical flow, excellent formatting and clarity

Note to students: Please attach this appendix together with the submission of coursework.